

Ismail Aissaoui
Email: i.aissaoui@esi-sba.dz
Phone: +213 660 70 77 96
Portfolio: ismail-aissaoui.vercel.app

Philippe Palanque (IRIT)

Subject: Application for PhD Position – Engineering Digital Twins Dedicated to 2D and 3D Input and Output Technologies

Dear Philippe Palanque,

I am applying for the PhD position on "Engineering Digital Twins Dedicated to 2D and 3D Input and Output Technologies" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

My experience engineering the 'Live Control Room'—a high-performance Next.js and Three.js 3D web dashboard—has given me direct insight into the complexities of human-in-the-loop visualizations. I have built architectures that seamlessly map complex simulation states into interactive 3D output technologies. I am highly motivated to further this research by formalizing interaction protocols and integrating diverse 2D/3D input devices directly into the digital twin loop.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui